

Module 1: Designing AI Solutions

Q1. What is stated as the primary overarching goal of designing AI solutions?

- A) Minimizing the parameters of neural networks.
- B) Ensuring systems deliver meaningful value by aligning with user expectations, organizational goals, and ethical considerations. **
- C) Replacing all human data engineering teams.
- D) Running automated code generation endpoints.

Q2. During user research phase of an AI project, which practice is explicitly recommended to represent audience segments?

- A) Quantization step metrics
- B) Persona Development **
- C) Learning rate optimization
- D) Replay Buffer Allocation

Q3. Which design thinking step focuses on brainstorming creative solutions and exploring diverse ideas?

- A) Define
- B) Empathize
- C) Prototype
- D) Ideate **

Q4. In a structured AI workflow inspired by Google's AI Guidebook, what step follows Data Preparation?

- A) Continuous Ingestion
- B) Model Development **
- C) Problem Understanding
- D) API Containerization

Q5. In the chatbot case study, by how much did the deployment of the transformer-based model reduce human agent workload?

- A) 10%
- B) 25%
- C) 40% **
- D) 75%

Module 2: AI Frameworks Overview

Q6. Why are AI frameworks critical to the modern development ecosystem?

- A) They completely replace the need for physical GPUs.
- B) They provide pre-built components and libraries that simplify development, ensuring scalability and performance optimization. **

- C) They convert Python scripts directly into hardware circuit boards.
- D) They eliminate the necessity of collecting training data.

Q7. Which open-source framework is highlighted as particularly famous for its dynamic computation graph and ease of research use?

- A) TensorFlow
- B) Scikit-learn
- C) PyTorch **
- D) Apache Spark Core

Q8. If your team is handling an unsupervised learning task like clustering or dimensionality reduction on standard data, which versatile Python library is most appropriate?

- A) OpenAI Gym
- B) Stable-Baselines3
- C) Scikit-learn **
- D) Flask REST API

Q9. Which tool provides a specialized benchmarking toolkit containing diverse simulation environments to develop and compare reinforcement learning agents?

- A) H2O.ai
- B) OpenAI Gym **
- C) MLflow
- D) PyCaret

Q10. Stable-Baselines3 offers state-of-the-art reinforcement learning algorithms built natively on top of which library?

- A) TensorFlow
- B) PyTorch **
- C) Apache Spark
- D) Hugging Face

Q11. Which module inside the Apache Spark ecosystem is specifically engineered to deliver scalable machine learning on massive big data sets?

- A) Spark SQL
- B) PyCaret Setup
- C) MLlib **
- D) Advanced LIME

Q12. What primary service advantage does Amazon ML offer for enterprise pipelines?

- A) It executes local Jupyter notebooks without internet connectivity.
- B) It simplifies model deployment within scalable cloud environments, integrating with other AWS services. **

- C) It translates deep learning weights directly into legal contract summaries.
- D) It automates local Docker installations.

Q13. In the provided hands-on TensorFlow image classification code snippet, what function is utilized to normalize pixel inputs?

- A) `layers.Flatten()`
- B) Dividing raw image arrays by `255.0 **`
- C) `model.compile(optimizer='adam')`
- D) `cifar10.load_data()`

Q14. Reviewing the hands-on image classification script, which activation function is applied to the final output layer to calculate multi-class probabilities?

- A) `relu`
- B) `tanh`
- C) `softmax **`
- D) `sigmoid`

Q15. What optimizer and loss function combination is compiled in the course's TensorFlow step-by-step example?

- A) Adam and sparse categorical cross-entropy **
- B) SGD and Mean Squared Error
- C) RMSprop and Hinge Loss
- D) Optuna and Binary Cross-Entropy

Module 3: AutoAI and AI Scaling

Q16. AutoAI workflows accelerate development by automating which sequence of engineering tasks?

- A) API Gateway authorization and cloud firewall billing.
- B) Data preprocessing, feature engineering, model selection, and hyperparameter tuning.**
- C) Physical sensor hardware soldering.
- D) Human stakeholder interviews.

Q17. PyCaret is fundamentally characterized as what style of machine learning tool?

- A) A high-resource distributed cluster operating framework.
- B) An open-source, low-code machine learning library for quick prototyping. **
- C) A black-box medical diagnostics visualizer.
- D) A prompt optimization script for large language models.

Q18. Which PyCaret function is explicitly called to automatically benchmark multiple algorithms and print their performance comparisons side-by-side?

- A) `setup()`

- B) `get_data()`
- C) `compare_models()` **
- D) `save_model()`

Q19. When datasets scale to sizes where single-machine memory architectures fail, how does AI Scaling resolve the constraint?

- A) By purging all non-integer rows from the database.
- B) By distributing data processing and model training tasks across multiple cluster nodes.**
- C) By switching deep learning networks back to simple linear regressions.
- D) By forcing models to train exclusively on edge devices.

Q20. In an Apache Spark machine learning pipeline, which module is utilized to assemble raw individual data columns into a single unified features vector format?

- A) `KMeans()`
- B) `VectorAssembler` **
- C) `LogisticRegression()`
- D) Spark Streaming API

Q21. In the telecommunications customer churn scale case study, what combination of tools was deployed to process initial exploration and subsequent big data scaling?

- A) OpenAI Gym followed by Stable-Baselines3
- B) PyCaret for initial exploration, transitioning to Apache Spark for distributed scaling**
- C) Hugging Face tokenizers running on local Flask servers
- D) Docker configurations managed by basic Jenkins scripts

Module 4: MLOps Deployment and Lifecycling

Q22. What is the explicit operational objective of MLOps?

- A) To eliminate the need for cloud infrastructure platforms.
- B) To unify machine learning system development and operations, ensuring scalable, reliable, and maintainable systems.**
- C) To design user research personas for marketing teams.
- D) To automate the collection of user surveys.

Q23. Which automated engineering branch focuses specifically on streamlining the flow of data between teams to preserve high-quality datasets?

- A) DevOps
- B) DataOps **
- C) ModelOps
- D) Prompt Engineering

Q24. Which infrastructure tool is highlighted for model containerization tasks inside a modern MLOps workflow?

- A) Jenkins
- B) Git
- C) Docker **
- D) Apache Airflow

Q25. What technology is explicitly listed for container orchestration roles within production DevOps/MLOps pipelines?

- A) Fast API
- B) Flask
- C) Kubernetes **
- D) Pandas

Q25. Which tool can be used to schedule and orchestrate automated workflows and data ingestion processes?

- A) Optuna
- B) Hyperopt
- C) Apache Airflow **
- D) ONNX

Q26. If an engineering team wants to monitor model drift over time, which tool is explicitly recommended by the course?

- A) Docker
- B) MLflow **
- C) Helm Charts
- D) Jupyter Notebooks

Q27. Why would an MLOps engineer convert a trained model into the ONNX format before production deployment?

- A) To automatically clean missing data values.
- B) To ensure cross-platform compatibility across diverse production hosting systems. **
- C) To reduce parameter scale down to a few thousand tokens.
- D) To satisfy legal GDPR data governance rules.

Q28. Which platform specializes in natural language processing (NLP), providing a rich ecosystem of pre-trained models, fine-tuning infrastructure, and streamlined deployment?

- A) Vertex AI
- B) Hugging Face **
- C) IBM AI Fairness 360
- D) Scikit-learn

Q29. Google Cloud's fully integrated framework enabling streamlined model management and enterprise deployment is known as:

- A) Azure ML
- B) Amazon SageMaker
- C) Vertex AI **
- D) Apache Spark SQL

Module 5: Generative AI Foundations

Q30. What marks the core functional difference between Generative AI models and traditional classification models?

- A) Generative AI models are resource-efficient and run exclusively without parameter weights.
- B) Generative AI focuses on synthesizing entirely new data instances that closely resemble the patterns of input data. **
- C) Generative AI models cannot process text.
- D) Generative AI relies on hard-coded rule scripts.